



Project Report

Bachelor of Computer Applications

Semester – IV

Sub: Data Warehousing and Mining

Topic: Marketing and Sales Analytics Dashboard

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March 2026

Project Title: Marketing And Sales Analytics Dashboard

1. ABSTRACT:

In the modern era of the digital commerce, Business generates enormous volumes of transactional data every day. However, raw data in spreadsheet format is difficult to interpret and even harder to act upon without proper visualization tools. This project addresses the challenge by building a fully interactive, multi-page Business Intelligence dashboard using Microsoft Power BI Desktop.

The data set used in this project is **Online Sales Dataset** sourced from Kaggle, An open-source data science platform. This data set contains 240 sales transaction recorded across six product categories spread across multiple global regions, it includes key attributes such as transaction date, product name, unit sold, unit price, total revenue, region and payment.

The primary problem addressed is a lack of centralized visual system to monitor sales performance, category trends and region distribution. Power BI was used as a primary tool for the data cleaning, modelling, DAX calculations and interactive dashboard development. The expected outcome is a fully interactive 5-page dashboard that enables business users to make a data driven decision effectively.

2. INTRODUCTION:

In today's data-driven world, organizations rely heavily on data analysis and visualization to make informed business decisions. Raw data alone is difficult to interpret, and therefore tools like Microsoft Power BI are used to convert complex datasets into meaningful and interactive visual insights. Data visualization helps in identifying patterns, trends, and relationships that support effective decision-making.

- **Background of the Problem:**

In today's competitive business environment, raw data alone is not efficient for decision making. Organizations need visual, interactive and real time dashboards to understand their sales performance. Customer behaviour and market trends. Business intelligence tools like Power BI bridge the gap between Raw Data and actionable insights.

- **Importance of visualization:**

Data visualization is the graphical presentation of data and information. When data is presented visually through graphs, charts, maps and dashboards. Human brain processes it significantly faster than when reading rows and columns of numbers. It transforms complex datasets into easy-to-understandable format. It allows stakeholders to identify patterns, track KPIs, and respond quickly to business changes without needing technical expertise.

- **Project Objective:**

The objective of this project is to build a comprehensive 5-page Power BI dashboard using the online sales dataset that covers Overview, Trend analysis. Category analysis Geographical insights, and a Detailed report with full interactivity including slicers, drill down, drill through, tool tips, buttons and bookmarks.

3. PROBLEM STATEMENT:

Online retail businesses generate thousands of transactions daily across multiple product categories and regions. Without a proper visualization system, it is difficult to:

- Track total revenue and transaction trends over time.
- Identify best performing and worst performing product categories.
- Understand which region contribute the most to overall sales.
- Monitor high-value vs low-value transactions.
- Make Quick business decision based on payment methods and product brands.

This project solves this problem by building an interactive Power BI Dashboard that provides a 360-degree view of business performance using the online sales dataset.

4. DATASET DESCRIPTION:

Dataset Source: Kaggle — Online Sales Dataset (Popular Marketplace Data) **Link:** <https://www.kaggle.com/datasets/shreyanshverma27/online-sales-dataset-popular-marketplace-data>

Number of Rows: 240 transactions **Number of Columns:** 9 columns (original) → 11 columns (after transformations)

Feature Description:

Column Name	Description	Data Type
Transaction ID	Unique identifier for each sale	Text
Date	Date of transaction	Date
Product Category	Category of product sold	Text
Product Name	Name of the product	Text
Units Sold	Number of units sold per transaction	Whole Number
Unit Price	Price per unit in USD	Decimal
Total Revenue	Total revenue = Units × Price	Decimal
Region	Geographic region of sale	Text
Payment Method	Mode of payment used	Text
Revenue Band	Categorized as High/Medium/Low	Text
Brand	Extracted from Product Name	Text

5. DATA PREPROCESSING (POWER QUERY): All data cleaning was performed inside Power Query Editor in Power BI Desktop.

Step 1 — Import Data: The CSV file was imported using Get Data → Text/CSV option in Power BI Desktop. Transform Data was selected to open Power Query Editor.

Step 2 — Remove Duplicates: The Transaction ID column was selected, and Remove Duplicates was applied to eliminate any repeated transaction records.

Step 3 — Handle Missing Values: Null values in the Region and Payment Method columns were replaced with "Unknown" using the Replace Values option. Blank rows were removed using Remove Rows → Remove Blank Rows.

Step 4 — Rename Columns: Unit Price was renamed to Price per Unit and Total Revenue was renamed to Revenue for better readability.

Step 5 — Change Data Types:

- Date column → Date type
- Revenue column → Decimal Number
- Units Sold column → Whole Number
- Transaction ID → Text

Step 6 — Filter Data: Rows with null Region values and zero Units Sold were filtered out to remove invalid or test records.

Step 7 — Split Columns: The Product Name column was split by delimiter (space) to extract Brand and Model as two separate columns using Split Column → By Delimiter.

Step 8 — Merge Tables: A duplicate query named Category_Lookup was created containing only the Product Category column. This was merged with the main Sales query using Left Outer Join on Product Category to simulate a Star Schema data model.

Step 9 — Add Conditional Column: A new column Revenue Band was created using Add Column → Conditional Column with the following rules:

- Revenue > 1000 → High
- Revenue > 500 → Medium
- Else → Low

Step 10 — Remove Unnecessary Columns: Auto-generated index columns and duplicate columns created during merge were removed. Close & Apply was clicked to load the cleaned data into Power BI.

6. DATA MODELING:

- **Tables Used:**
 - **Sales** (Main Fact Table) — Contains all 11 cleaned columns
 - **Category_Lookup** (Dimension Table) — Contains Product Category
- **Relationships Created:**
 - A One-to-Many relationship was created between:
Category_Lookup[Product Category] → Sales[Product Category]
- **Schema Explanation:** The data model follows a **Star Schema** design where the Sales table acts as the central fact table and Category_Lookup acts as a dimension table. This design improves query performance and allows cleaner DAX calculations. **This uses one to many Relationship**

Category_Lookup (Secondary Table)

[Product Category]



Sales (Main Table)

[Transaction ID, Date, Brand, Model,

Product Category, Units Sold, Price per Unit,

Revenue, Region, Payment Method, Revenue Band]

7. DAX CALCULATIONS:

DAX (Data Analysis Expressions) was used to create measures for analysing the dataset. All DAX measures were stored in a dedicated _Measures table for professional organization.

Measures Created:

1. **Total Revenue:** Total Revenue = SUM (Sales [Revenue]). It calculates the total sales revenue across all transactions.
2. **Average Order Value:** Avg Order Value = AVERAGE (Sales [Revenue]). It calculates the average revenue per transaction.
3. **Total Transactions:** Total Transactions = COUNT (Sales [Transaction ID]). It counts the total number of sales transactions.
4. **Unique Products:** Unique Products = DISTINCTCOUNT (Sales [Model]). It counts the number of distinct products sold.
5. **Revenue Status:** Revenue Status = IF (SUM (Sales [Revenue]) > 50000, "Above Target", "Below Target"). It flags overall revenue performance against target.

6. Electronics Revenue: Electronics Revenue = CALCULATE (SUM (Sales [Revenue]), Sales [Product Category] = "Electronics"). It calculates revenue specifically for Electronics category.
7. High Value Transactions: High Value Transactions = CALCULATE (COUNT (Sales [Transaction ID]), FILTER (Sales, Sales [Revenue] > 500)). It counts transactions with revenue above \$500.
8. Revenue % of Total: All Revenue = CALCULATE (SUM (Sales [Revenue]), ALL(Sales)) Revenue % of Total = DIVIDE ([Total Revenue], [All Revenue], 0). It shows each category's percentage contribution to total revenue.
9. Highest Transaction: Highest Transaction = MAX (Sales [Revenue]). It returns the highest single transaction value.
10. Lowest Transaction: Lowest Transaction = MIN (Sales [Revenue]). It returns the lowest single transaction value.

8. DASHBOARD DEVELOPMENT:

1. Overview Dashboard:

- Charts Used: 4 KPI Cards, Donut Chart, Clustered Bar Chart, Line Chart
- Slicers: Date Range Slicer
- Purpose: Provides a high-level summary of Total Revenue (\$80.57K), Total Transactions (240), Avg Order Value, and Unique Products. Shows revenue by category and top brands.

2. Trend Analysis:

- Charts Used: Line Chart with Date Hierarchy, Area Chart, Matrix
- Interactivity: Drill-down enabled (Year → Quarter → Month → Day)
- Purpose: Tracks revenue and units sold trends over time. Matrix shows Month vs Category revenue breakdown.

3. Category Analysis:

- Charts Used: Clustered Bar Chart, Treemap, Pie Chart, Summary Table
- Slicers: Product Category Dropdown Slicer
- Interactivity: Drill-through to Detailed Report, Bookmarks for filtered views
- Purpose: Compares performance across 6 product categories — Electronics, Clothing, Sports, Books, Beauty Products, Home Appliances.

4. Geographical Report:

- Charts Used: Filled Map, Clustered Bar Chart, Region Summary Table
- Slicers: Region List Slicer
- Purpose: Visualizes revenue distribution across different regions using bubble size on map and bar chart rankings.

5. Detailed Report:

- Charts Used: Full Data Table
- Slicers: Date Range, Product Category, Region, Payment Method, Revenue Band
- Interactivity: Drill-through target page — receives filtered data from other pages
- Purpose: Shows granular transaction-level data for deep analysis.

6. Interactivity Features Implemented:

- Slicers — Date, Category, Region, Payment Method on respective pages
- Drill-down — Date hierarchy on Line Chart (Year → Month → Day)
- Drill-through — From Category Analysis → Detailed Report filtered by category
- Tooltips — Custom tooltip page showing KPIs on hover
- Buttons — Navigation buttons on Overview Dashboard to jump between pages
- Bookmarks — Saved views for Electronics, All Categories, and High Revenue filters

9. KEY INSIGHTS: Based on the structure and measures in the dashboard, the following business insights can be derived:

- **Revenue concentration:** The Donut Chart and Category Analysis page reveal which product categories contribute the dominant share of total revenue — useful for prioritizing inventory and marketing investment.
- **Seasonal trends:** The line and area charts on the Trend Analysis page expose peaks and troughs in revenue and units sold across months and quarters — critical for demand forecasting.
- **Brand performance:** The clustered bar chart on the Overview Dashboard highlights which brands are top revenue generators, informing partner/vendor strategy.
- **Regional gaps:** The map and bar chart on the Geographical Report expose high and low-performing regions, enabling targeted sales campaigns.
- **Payment method analysis:** The Detailed Report's slicer on Payment Method allows filtering to understand which payment channels are most used — useful for checkout optimization.
- **Revenue banding:** The Revenue Band slicer segments transactions into value tiers, helping identify whether revenue is driven by many small orders or fewer high-value ones.

10. SCREENSHOTS:

Power Query steps:

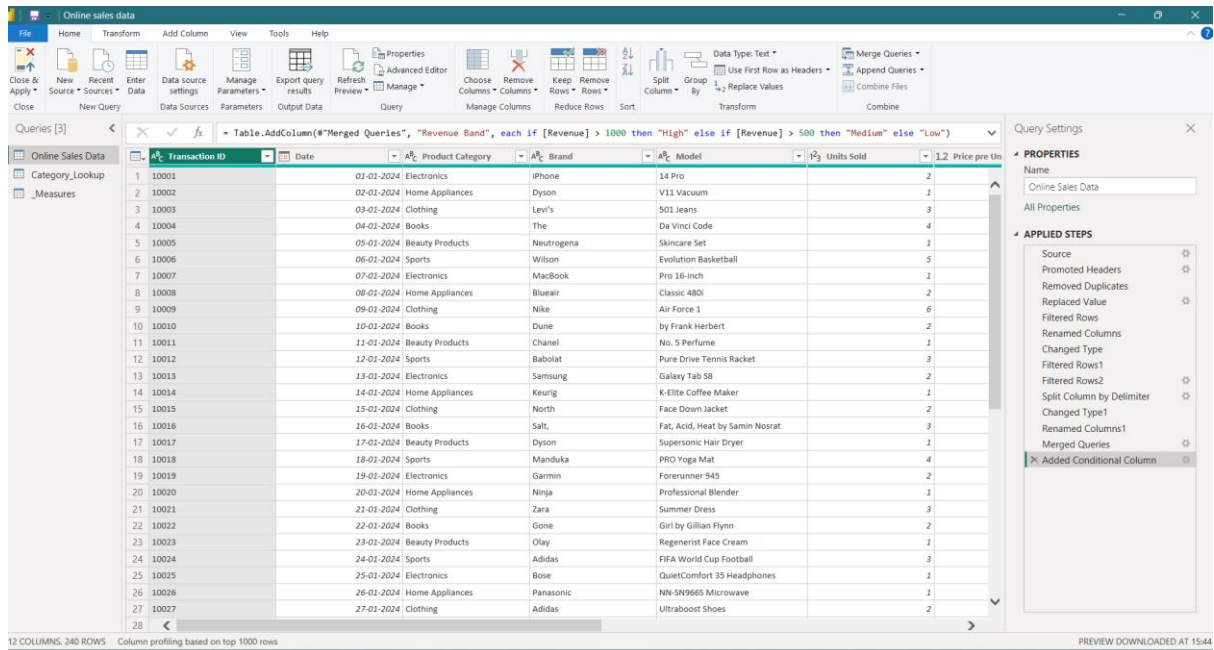


Figure: Power Query Editor showing data cleaning and transformation steps

Data Model:

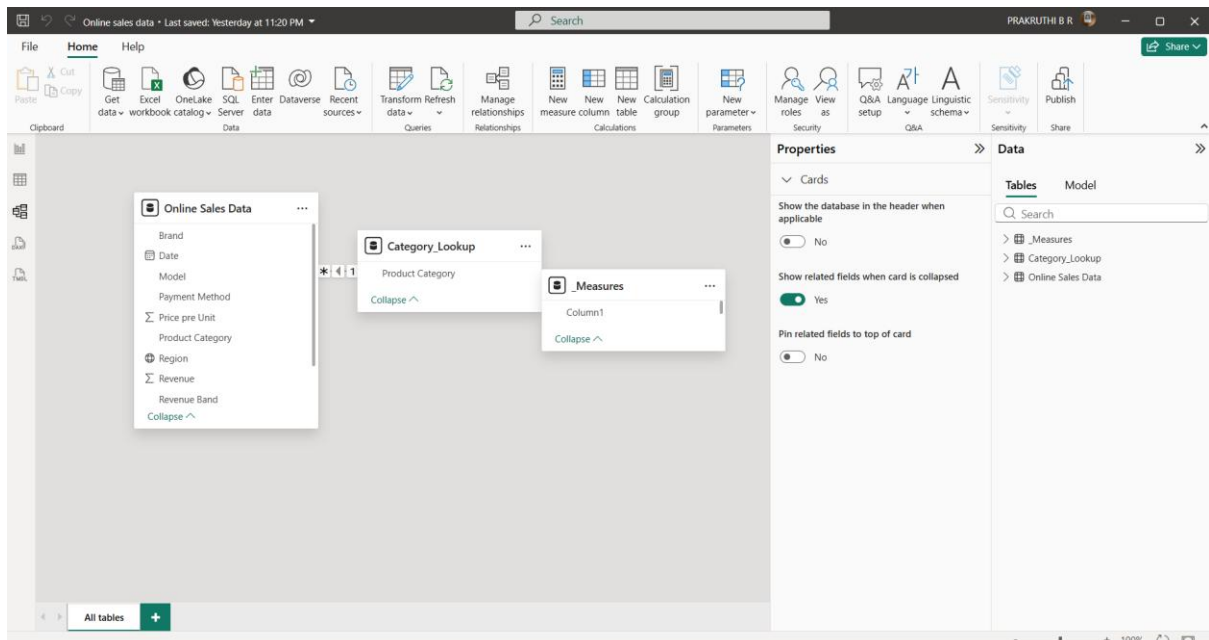
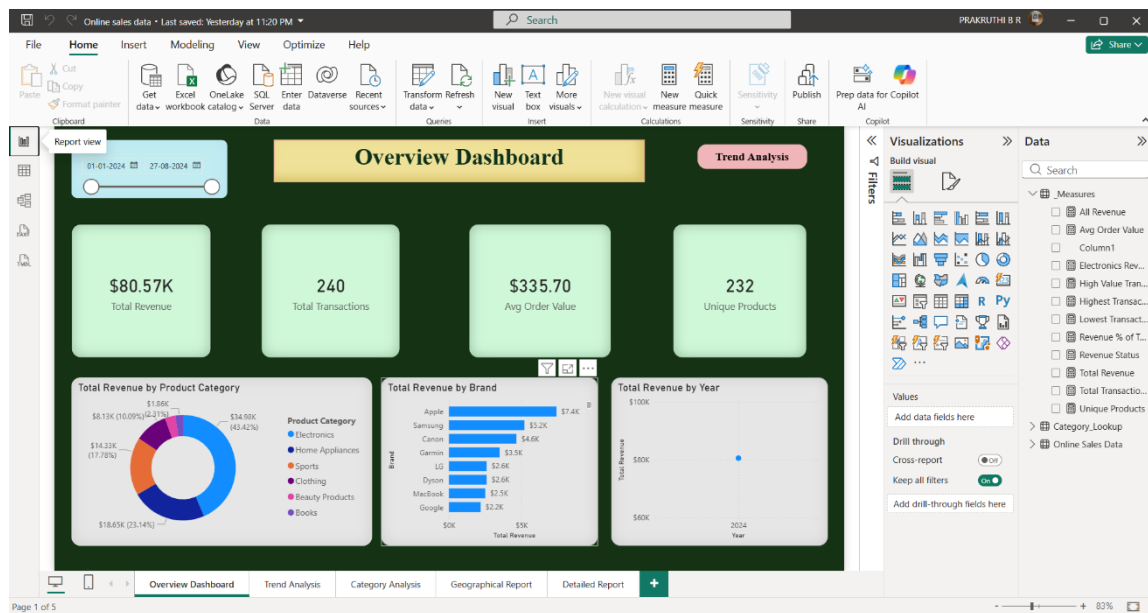


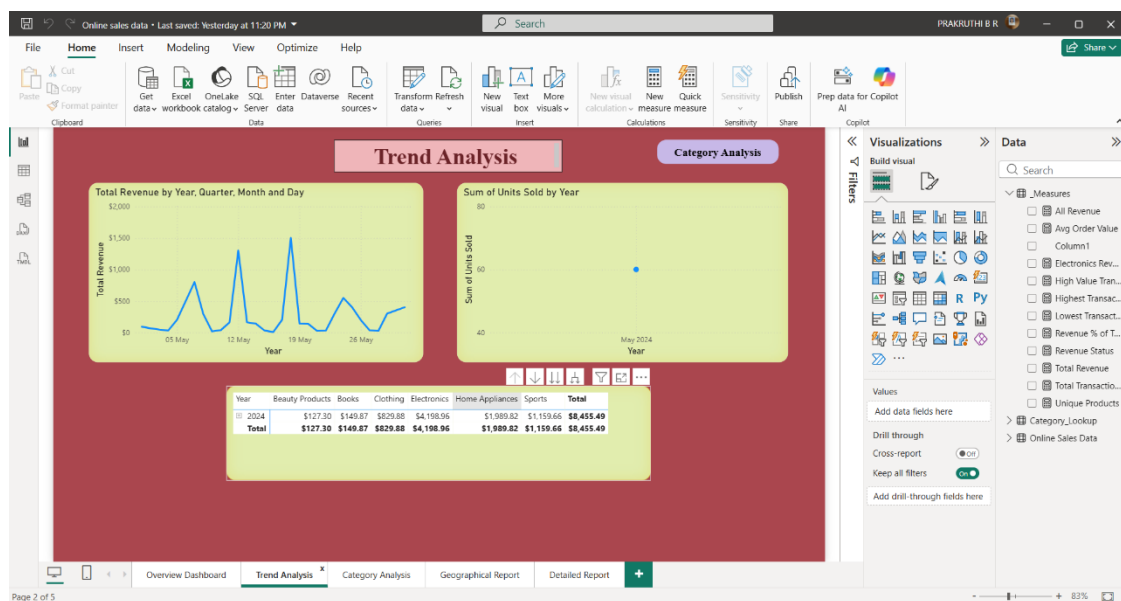
Figure: Data Model view showing the structure of the dataset in Power BI

Dashboard: The dashboard was developed using Power BI to provide an interactive and user-friendly interface for data analysis. Multiple report pages were created to analyse the dataset from different perspectives.

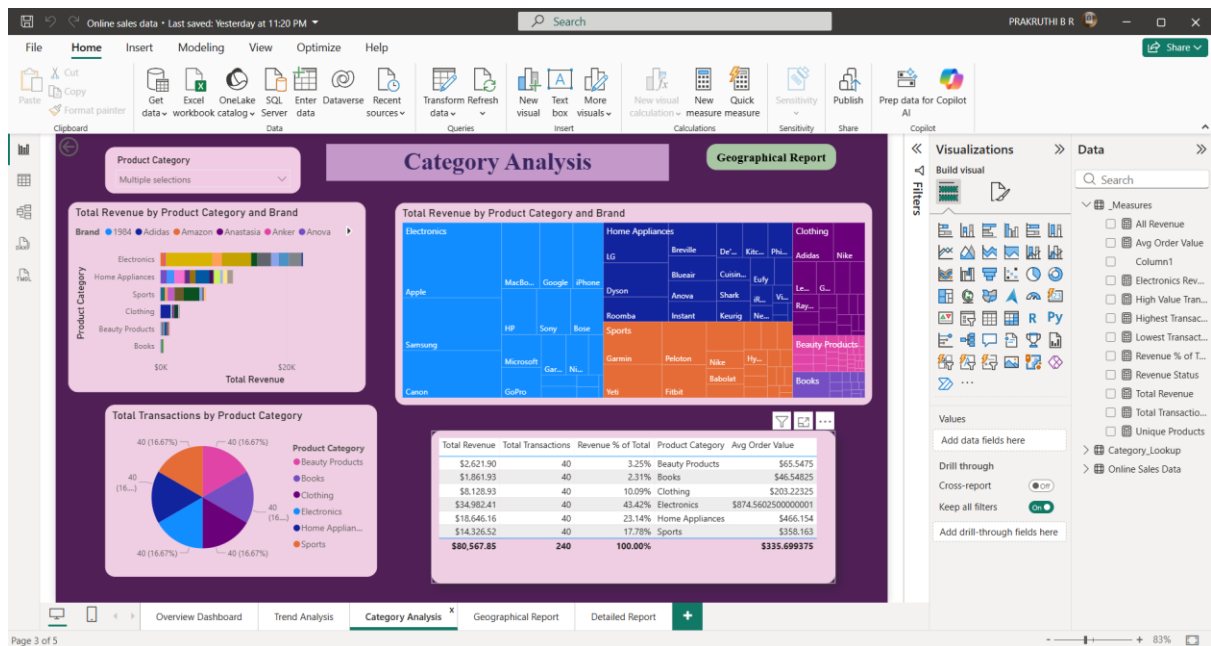
- **Overview Dashboard:** It provides a summary of key metrics like total revenue, transactions, average order value, and product count. It gives a quick view of overall performance and highlights major trends.



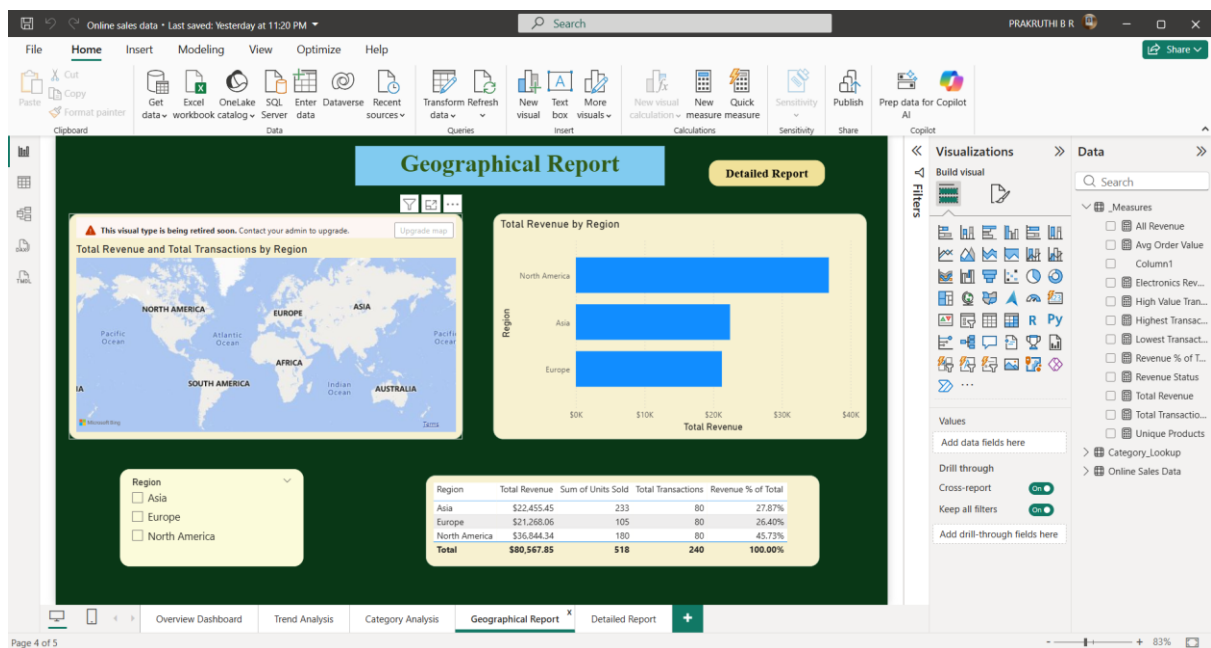
- **Trend Analysis:** Shows how revenue and sales change over time, with noticeable fluctuations and peak periods. Helps identify patterns, growth trends, and periods of high or low performance.



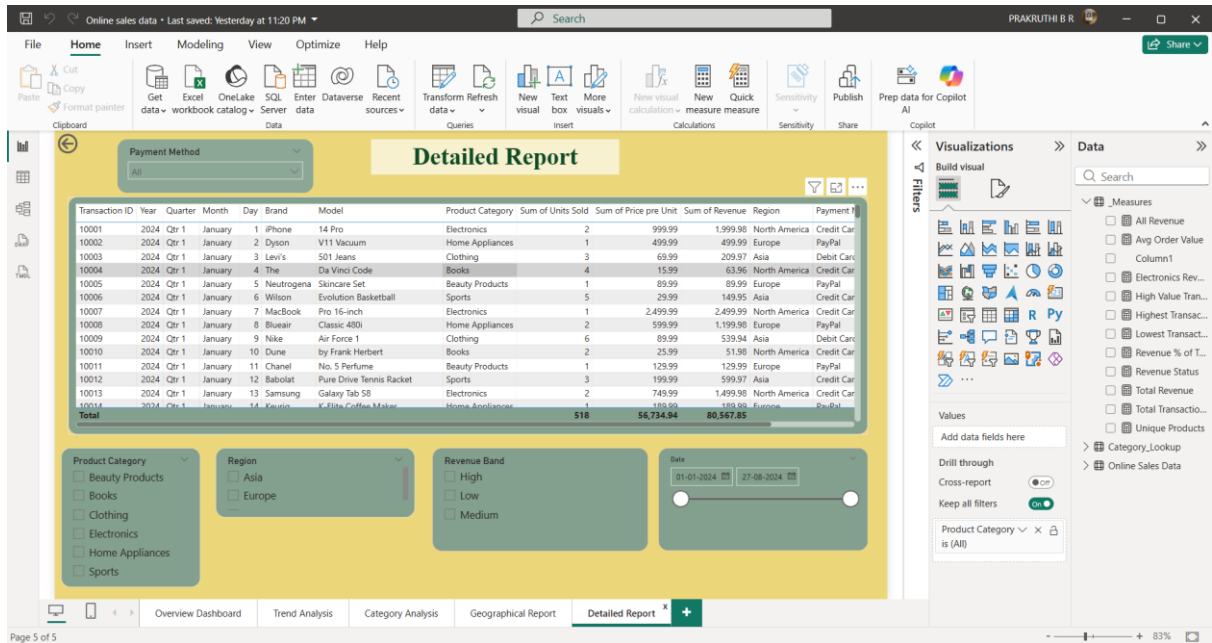
- **Category Analysis:** Displays revenue distribution across product categories and brands using charts. Electronics is the top-performing category, while Books and Beauty Products contribute the least.



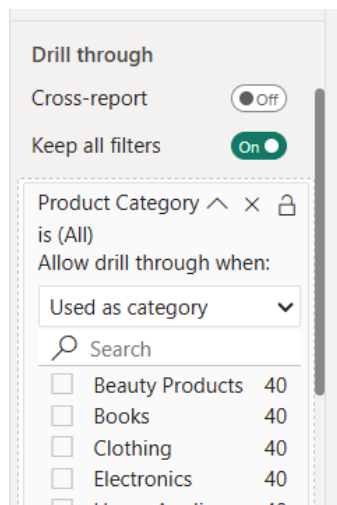
- **Geographical Report:** Visualizes sales performance across different locations using map-based visuals. Helps identify regions with high and low revenue contributions.



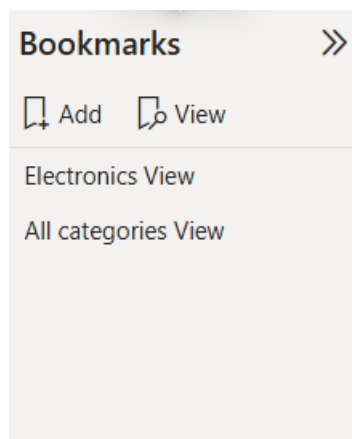
- **Detailed Report:** Provides a detailed breakdown of data in tabular format for deeper analysis. Allows users to explore specific values, categories, and performance metrics.



- **Drill-through:**



- **Bookmarks Panel:**



11. CHALLENGES FACED:

- **Map Visual Disabled** — Power BI had map visuals disabled by default. Fixed by enabling them in File → Options → Security → Use Map and Filled Map visuals.
- **Sign-in Popup Interruption** — Power BI repeatedly showed "Enter your email address" popup while working. Resolved by signing in with a Microsoft account once.
- **Drill-through Not Working** — Drill-through navigation was not jumping to Detailed Report page. Fixed by holding Ctrl key while right-clicking in Edit mode and ensuring Product Category column (not measure) was placed in the Drill-through field.
- **Color Saturation Field Missing** — The Filled Map field "Color Saturation" was renamed to "Bubble Size" in newer Power BI versions. Resolved by using Bubble Size field instead.
- **Column Name Mismatch** — Order ID column referenced in guides did not exist in dataset. Used Transaction ID as the correct unique identifier column.
- **Data Type Errors** — Some columns had incorrect auto-detected data types causing DAX errors. Fixed by manually setting correct data types in Power Query Editor.

12. CONCLUSION:

This project successfully demonstrates the end-to-end development of a Business Intelligence solution using Microsoft Power BI. Starting from raw CSV data downloaded from Kaggle, the project covered all major stages including data import, cleaning using Power Query (10 transformations), data modelling with Star Schema, DAX measure creation (10 measures), and development of 5 interactive report pages. The final dashboard provides comprehensive insights into sales performance, category trends, regional distribution, and transaction-level details. The interactivity features including slicers, drill-down, drill-through, tooltips, buttons, and bookmarks make the dashboard user-friendly and suitable for real business use cases.

13. FUTURE ENHANCEMENTS:

1. **Real-time Data Integration** — Connect Power BI to a live database or API to enable real-time dashboard updates without manual CSV refresh
2. **AI-powered Insights** — Use Power BI's built-in AI visuals like Key Influencers and Decomposition Tree for automated insight generation
3. **Forecasting** — Add time-series forecasting using Power BI's Analytics pane to predict future revenue trends

4. **Row-Level Security (RLS)** — Implement RLS so different managers only see data for their own regions
5. **Power BI Service Publishing** — Publish the dashboard to Power BI Service for cloud access and scheduled data refresh
6. **Mobile Layout** — Design a mobile-optimized layout using Power BI Mobile view for on-the-go access
7. **Customer Segmentation** — Integrate RFM (Recency, Frequency, Monetary) analysis for advanced customer segmentation

14. REFERENCES:

1. **Dataset:** Online Sales Dataset — Popular Marketplace Data
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3. **DAX Reference:** Microsoft DAX Function Reference <https://docs.microsoft.com/en-us/dax/>
4. **Power Query Reference:** Microsoft Power Query Documentation
<https://docs.microsoft.com/en-us/power-query/>
5. **Kaggle Platform:** <https://www.kaggle.com>